

O. I. Craciunescu
email: oana@radonc.duke.edu

S. K. Das
email: shiva@radonc.duke.edu

Department of Radiation Oncology,
Duke University Medical Center,
Box 3455,
Durham, NC, 27710

S. T. Clegg
Ceprobe Corporation,
Gilbert, AZ 85233
email: sclegg@mail.crbp.org

Dynamic Contrast-Enhanced MRI and Fractal Characteristics of Percolation Clusters in Two-Dimensional Tumor Blood Perfusion

Dynamic contrast-enhanced magnetic resonance imaging (DE-MRI) of the tumor blood pool is used to study tumor tissue perfusion. The results are then analyzed using percolation models. Percolation cluster geometry is depicted using the wash-in component of MRI contrast signal intensity. Fractal characteristics are determined for each two-dimensional cluster. The invasion percolation model is used to describe the evolution of the tumor perfusion front. Although tumor perfusion can be depicted rigorously only in three dimensions, two-dimensional cases are used to validate the methodology. It is concluded that the blood perfusion in a two-dimensional tumor vessel network has a fractal structure and that the evolution of the perfusion front can be characterized using invasion percolation. For all the cases studied, the front starts to grow from the periphery of the tumor (where the feeding vessel was assumed to lie) and continues to grow toward the center of the tumor, accounting for the well-documented perfused periphery and necrotic core of the tumor tissue.

Introduction

It is well known that the delivery of local and regional cancer hyperthermia therapy is dependent upon tumor vascularization: Hyperthermia is more effective in poorly vascularized regions of malignant tissue (Dewhirst et al., 1997; Samulski and Fessenden, 1990). Thus, knowledge of blood perfusion distribution is necessary in order to optimize cancer hyperthermia treatments. Perfusion data require an accurate description of the tumor vascular network. Models for blood perfusion in normal or tumor tissue have been used in the literature. An up-to-date description of the efforts to model perfusion is given elsewhere (Craciunescu, 1998). Unfortunately, tumor vessels are tortuous and sinuous, have increased vascular length and diameter compared to normal tissue, shunts, loops, variable intravascular distances, and the whole structure is heterogeneous (Dewhirst, 1993; Warren, 1979). These characteristics result in biological structures that cannot be easily described using deterministic models. A deterministic model for the tumor vascular network might be possible, but very laborious, because the system to be described is very complex. Angiogenesis in tumors encompasses a distribution of patterns with a non-analytical characteristic function. For a deterministic model, we need the distribution of the vessel lengths, diameters, and curvatures to determine the flow field. But this might not be enough. It is known that the physiology of the tumor vessel walls is completely different than for normal vessels. **In tumors, due to the permeability of the high vessel walls, the blood flow is not restricted to the vascular network, but occurs in the interstitial space as well.**

As an alternative, the irregular and fragmented patterns of angiogenesis and perfusion have been described using fractal geometry (Mandelbrot, 1982; Tsonis and Tsonis, 1987; Kiani and Hudetz, 1991; Vico et al., 1992; Gazit et al., 1995, 1997).

In particular, percolation theory has been used to describe vascular networks (Gazit et al., 1995, 1997; Daxter and Ettl, 1995; Hudetz, 1993; Daxter, 1993; Hudetz and Werin, 1986). Baish et al. (1996) used the invasion percolation network model to determine the role of tumor microvascular architecture in drug delivery. The first to show that tumors display "percolation-like scaling" by studying the scale-invariant behavior of normal and tumor microvascular networks were Gazit et al. (1995, 1997). Their study used skeletonized images of tumor microvascular networks grown in mice bearing dorsal skinfold chambers to observe the tumor microvascular geometry. From their observations, they were able to conclude that the tumor vasculature could be described using invasion percolation. However, from the hyperthermia point of view, most bioheat transfer takes place not at the microvascular level, but in vessels larger than those previously described using fractal analysis, hence the whole blood pool must be considered.

The introduction of complex internal geometries into flow problems greatly increases the required physical information and mathematical complexity. The purpose is to find morphological properties that are useful to construct a reductive model for the transport problem. Percolation theory is a mathematical tool that allows the prediction of morphological and transport properties for heterogeneous materials by the use of simple scaling laws. Assuming that tumor tissue is a porous medium, subdividing it into many small spaces and labeling the occupant of each subdivision as "conductive" (with flow) or "nonconductive" (no flow), it is then possible to use percolation theory to describe the transport and geometric properties of the material. A continuous pathway of conductive sites that span the tumor is formed and transport takes place through the tumor.

Dynamic contrast-enhanced magnetic resonance imaging (DE-MRI) of spontaneous canine tumors is used in this paper to measure the tumor perfusion distribution. The tumor perfusion is considered to take place in a percolation-like cluster. The cluster geometry is depicted using the contrast wash-in component (a surrogate for blood perfusion). Fractal characteristics are determined for each two-dimensional cluster. Invasion percolation, as a

Contributed by the Bioengineering Division for publication in the JOURNAL OF BIOMECHANICAL ENGINEERING. Manuscript received by the Bioengineering Division August 11, 1998; revised manuscript received April 20, 1999. Associate Technical Editor: R. B. Roemer.

statistical growth process, is then used to monitor the perfusion front in the tumor. It is concluded that the blood perfusion in a two-dimensional tumor vessel network has a fractal structure and that the evolution of the perfusion front can be characterized using invasion percolation. This result allows the prediction of the perfusion front velocity by employing a simple scaling law characteristic to percolation models (Herrmann and Stanley, 1988). The local velocity of the perfusion front can be used to assess the perfusion term in the bioheat equation (Pennes, 1948), and thus predict the temperature distributions obtained during hyperthermia treatments. In addition, local fractal dimensions of tumor fragments might be correlated with the respective fragment hypoxic ratio, thus leading to a noninvasive mean of determining the tumor oxygenation.

Theoretical Background: Fractal Analysis

A space object or a time sequence is called fractal if it possesses two properties: self-similarity and scaling. A geometrically self-similar object is one that looks the same under different length scales. For scaling, the measured properties of an object have to depend on the scale at which they are measured (Bunde and Havlin, 1991). The "property"—length relation of such a structure is characterized by a power law with a non-integer exponent called the fractal dimension (Feder, 1988; Vicsek, 1989; Bunde and Havlin, 1991). This dimension tells us how many new pieces similar to the whole object are observed as the resolution is made finer. Objects with self-similarity dimensions of 1, 2, and 3 correspond to the usual ideas that one-dimensional objects occupy lines, two-dimensional objects occupy areas, and three-dimensional objects occupy volumes.

The typical length up to which the structure is self-similar and can be regarded as fractal is called the correlation length ξ (Bunde and Havlin, 1991). On length scales larger than ξ the clusters fill up the space homogeneously. The fractal dimension alone is not sufficient to describe a fractal structure completely (Stanley, 1991). Of importance in the study of fractal structures is the fractal dimension of the minimum path, $d_{\min}$, which governs the fashion in which information propagates from one point to another. In our case, $d_{\min}$ can be used to describe the tortuosity of the vessels in the network (Gazit et al., 1995, 1997) and also the velocity with which the perfusion front propagates in the tumor vascular network. The fractal dimension of the minimum path is determined from the power law $l_c = r^{d_{\min}}$, with l_c the chemical distance (the shortest path between two sites in the lattice), and r the Pythagorean distance between the two sites. The concept of chemical distance was shown to play an important role in the description of spreading phenomena, for example epidemics and forest fires, which propagate along the shortest path from the seed (i.e., from the starting point of the spreading phenomenon) (Bunde and Havlin, 1991). The tumor blood perfusion is such a spreading phenomenon and therefore presumably well described by $d_{\min}$.

Percolation is the simplest model of disordered systems that has fractal structure (Stauffer and Aharony, 1992; Bunde and Havlin, 1991). Percolation clusters encompass different substructures, each defined by its own fractal dimension. One such substructure relevant to the transport process across the percolation cluster is the backbone. The backbone is obtained by removing all dangling ends, i.e., branches that are not necessary for maintaining the connectivity of the structure throughout the lattice (Mastorakos and Argyrakakis, 1993).

Materials and Methods

Dynamic contrast-enhanced MRI is used to provide perfusion distribution data in six soft tissue sarcomas¹ located either in the legs or paws of six canine subjects.² For each case, the images used represented regions of interest selected to contain the tumor. The principle of DE-MRI for tumor blood perfusion measurements is described in detail by Belfi et al. (1992, 1994). The tumors were imaged using dynamic Gd-enhanced MRI (1.5 T Signa: GE Medical Systems, Milwaukee, WI). The MR protocol consisted of serial T1-weighted spin echo images before and after the injection of 0.15 mmol/kg of Gd-DOTA.³ Contrast enhancement was used for an increase in MR intensity, better specificity, better blood pool versus tissue characterization and reduction of image artifacts (Rinck, 1993). Typical pulse sequence acquisition parameters were: T_r (repetition time) = 400 ms, T_e (echo time) = 14–16 ms, FOV (field-of-view) = 100–160 mm, slice thickness = 3 or 4 mm, BW (full bandwidth) = ± 16 kHz, acquisition matrix = 256×256 , pixel size = 0.39–0.625 mm, N_x (number of excitations) = 1, and 8–15 slices.

The image intensity was processed on a pixel-by-pixel basis. The lower limit of the accessible length scales obtainable with MRI varied between $0.39 \times 0.39 \text{ mm}^2$ and $0.625 \times 0.625 \text{ mm}^2$ from case to case. Time versus signal intensity curves were fit using a double exponential (Kurland et al., 1989):

$$S_p = S_{\max} [1 - e^{r_1(t-t_0)}] e^{-r_2(t-t_0)} \quad (1)$$

where S_p is the percent image intensity enhancement⁴ (the relative difference between the post- and pre-injection images), $S_{\max}$ is the maximum image enhancement, r_1 is the rate of contrast wash-in, r_2 is the rate of contrast wash-out, t_0 is the time at the beginning of the Gd injection, and t is the current time. The wash-in rate monitors the change in the tissue image intensity, which in turn is proportional to the tumor blood perfusion (Belfi et al., 1992). The values for the wash-in are dependent on the region of interest (ROI) selected and the slice thickness.

¹ Two fibrosarcomas, two hemangiopericytomas, one myxosarcoma, and one neurofibroma.

² Subject's weight varied between 11 and 39 kg.

³ Prohance, Bracco.

⁴ $S_p = (I_{\text{post}} - I_{\text{pre}}) / I_{\text{pre}} \times 100$

Nomenclature

d_E = Euclidean dimension	R^2 = goodness of fit	t_0 = time at the beginning of contrast injection
d_f = fractal dimension of the percolation cluster	RPI = relative perfusion index	T_e = echo time
$d_{\min}$ = fractal dimension of the minimum path	$S_{\max}$ = maximum image enhancement	T_r = repetition time
I = image intensity	S_p = percent image enhancement	$\rho(L)$ = average cluster density
L = length scale	p = cluster concentration	ξ = correlation length
l_c = chemical distance	p_c = percolation threshold (critical concentration)	$(\cdot)_B$ = characteristics for the backbone of the percolation cluster
$M(L)$ = mass of the cluster	r = distance along the cluster	$(\cdot)_{\text{pre}}$ = pre-injection
N_x = number of excitations	r_1 = rate of contrast wash-in	$(\cdot)_{\text{post}}$ = post-injection
p_{∞} = probability that a site belongs to the percolation cluster	r_2 = rate of contrast wash-out	
	t = time	

A percolation model on a square (two-dimensional) lattice was applied to study the MRI contrast uptake and its fractal structure. The hypothesis that MR data can be represented using percolation theory is tested with two separate studies. In study one, percolation clusters are constructed from the contrast uptake data; and the cluster fractal characteristics (p , d_f , ξ , and d_{min}) are then computed and compared to known values for two-dimensional cross sections of three-dimensional percolation clusters. In the second study, the invasion percolation growth model is used to describe the dynamics of the perfusion front.

Study One. The wash-in images are a result of the motion of the contrast agent (Gd-DOTA) in the interstitial space. The motion can be due to either convection or diffusion. The clusters were determined in the following manner. First, the maximum value of the wash-in was identified. Then, the minimum value was chosen based upon the knowledge that any wash-in lower than the minimum value must be diffusion, not perfusion. By estimating the time it takes a Gd-DOTA molecule to move across a pixel given its diffusion length scale, we chose to set the minimum value of the wash-in data as 1 percent of the maximum value. All pixels with the wash-in less than the minimum value were considered unoccupied, the others were considered occupied. The structure thus obtained is similar to one obtained with site percolation. In order to retain only the percolating cluster (the structure that allows the flow to reach from one side of the lattice to the other), the Hoshen-Kopelman algorithm was used (Hoshen and Kopelman, 1976; Binder and Heermann, 1992). This algorithm labels all sites in the percolation system such that sites with the same label belong to the same cluster and different labels belong to different clusters. If the same label is found at opposite sides of the lattice, an infinite percolating cluster exists (Binder and Heermann, 1992).

The percolating cluster characteristics are then determined as follows: for each cluster an average density is determined as $\rho(L) = M(L)/L^{d_E}$, with $d_E = 2$ for two dimensions. $M(L)$ is the mass of the cluster and it was computed using the sandbox method (Bunde and Havlin, 1994), and L is the length scale. A plot of $\ln(\rho(L))$ against $\ln(L)$ exhibits three distinct regions, typical for percolating clusters (Kapitulnik et al., 1983): Region I follows a power law, $\rho(L) = r^{d_f - d_E}$; region II is independent of L and identified as p_∞ (the probability that an occupied site belongs to the infinite cluster), and region III shows boundary effects. The intersection of the least-squares-fit lines defining regions I and II yields the crossover point where $L = \xi$ (Kapitulnik et al., 1983). The concentration p of each of the clusters is estimated as the ratio of occupied versus total number of sites in the lattice.

The fractal dimension of the minimum path d_{min} is then computed using the power law $l_c = r^{d_{min}}$. The chemical distance l_c is determined using the algorithm proposed by Herrmann and Stanley (1988).

As a numerical/algorithmic check, the algorithms for computing the fractal dimensions (both d_f and d_{min}) were tested on computer-generated percolation clusters at the percolation threshold. The values obtained for the algorithmic test were: $d_f = 1.892 \pm 0.003$ and $d_{min} = 1.128$. These results are in good agreement with the values cited in the literature for two-dimensional site percolation clusters at the percolation threshold $p_c = 0.59277$ (Gebele, 1984): $d_f = 1.89583$ and $d_{min} = 1.13 \pm 0.002$ (Herrmann and Stanley, 1988).

Due to the construction of the percolation clusters from DE-MRI data, i.e., using wash-in values, the clusters are formed from sites with existing flow. Since the blood flow is continuous, the sites forming the perfusion cluster are connected. Therefore, we hypothesize that the infinite percolation cluster should be equivalent to the percolation backbone. For the clusters considered in this study, the backbone was obtained using the "burning" algorithm (Herrmann et al., 1984; Mastorakos and Argyrakakis, 1993). Then the backbone was compared to the original spanning cluster to test if they are indeed equivalent. The fractal dimension of the backbone was computed as described above for the whole cluster.

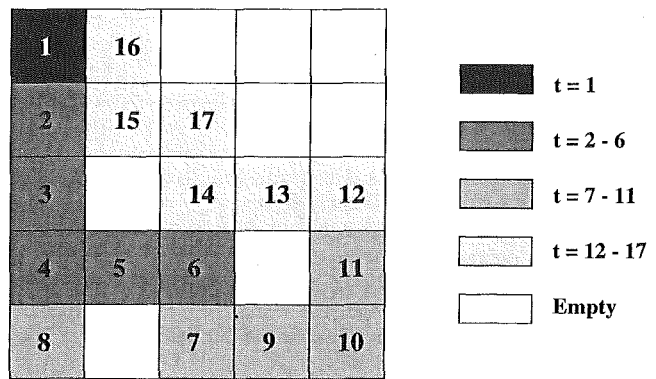


Fig. 1 Sample case used to describe the growth by invasion percolation model. The shades of gray represent the grown cluster at different times t .

Study Two. Invasion percolation involves the concept of the displacement of one fluid ("defender") by another ("invader"). The invader follows the dynamic rule of advancing its interface along the path of least resistance, as opposed to standard percolation where all interfaces were advanced up to some chosen threshold resistance (p) (Wilkinson and Willemsen, 1983).

For each image obtained from contrast enhanced MRI, a pixel with the highest flow rate was identified (highest $RPI = r_1 S_{max}$ value) as the source point (seed) of the cluster. Anatomically, we assume that this source is the main feeding vessel of the tumor under consideration. In order to grow the cluster in time (to follow the perfusion front) the invasion percolation growth model (IPGM) was applied. A simple IPGM on a given lattice is the following (Wilkinson and Barsony, 1984; McCarthy, 1987); see Fig. 1 for a sample case used to describe the growth by invasion percolation:

1. Identify a site to be the seed of the cluster.
2. Assign a number between 0 and 1 to the nearest-neighbor sites of the seed. The assigned number is based upon the recorded wash-in data that were scaled between 0 and 1. These sites form the boundary of the starting cluster.
3. Find the site on the boundary that has the largest wash-in value assigned to it.
4. Accept that site into the cluster and assign numbers between 0 and 1 to any of its nearest-neighbors that are empty, thereby adding them onto the boundary of the cluster.
5. Go to step (3).

The cluster growth is terminated if either the cluster is completely surrounded by zero pixels (and, therefore, cannot continue to grow) or after a predetermined number of iterations. In order to be able to record the evolution of the perfusion front in time, the clusters were grown at different percentages from the total growing time (total number of iterations). For all the intermediary clusters obtained, the fractal characteristics (p , d_f , ξ , and d_{min}) were determined using the same procedure as described for Study One.

Results

Study One. Wash-in data from different planes of six canine tumors were used to perform this study, i.e., a total of 20 configurations. Figure 2(a) shows a sample wash-in image obtained from contrast-enhanced MRI. The image represents the ROI that contains the tumor. Figure 2(b) shows intensity-time curves for three ROIs containing the tumor presented in Fig. 2(a) taken from a series of T_1 -weighted MR images ($T_r = 400$ ms, $T_e = 15$ ms), pre- and post-injection of 0.15 mmol/kg Gd-DOTA. The three ROIs represent, respectively, the tumor periphery, the whole tumor, and the tumor core. The pre-contrast images are at $t = 0$. The

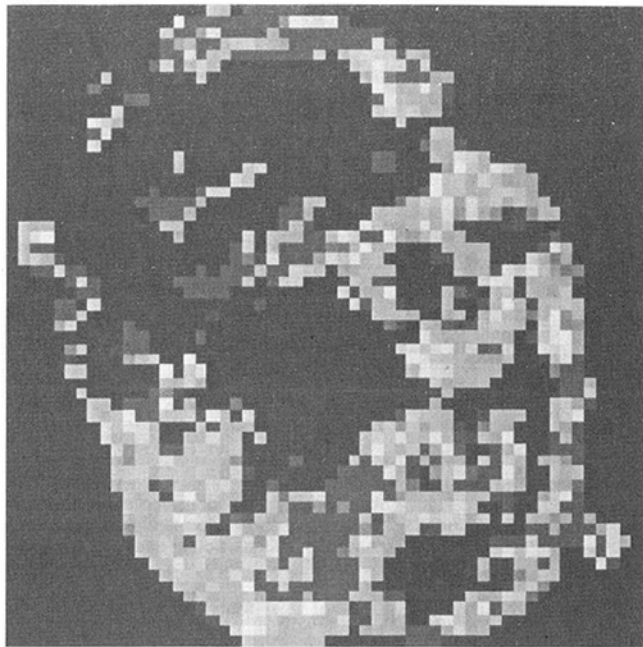


Fig. 2(a) Wash-in image of a midplane canine tumor obtained with dynamic contrast-enhanced MRI. For this particular case, the tumor is a hemangiopericytoma located on the right front paw of a canine weighing 11 kg.

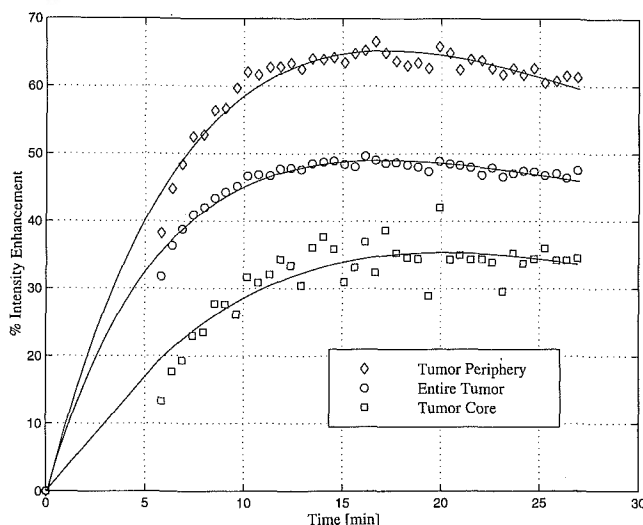


Fig. 2(b) Intensity-time curves for three ROIs containing the tumor presented in Fig. 2(a) taken from a series of T_1 -weighted MR images ($T_r = 400$ ms, $T_e = 15$ ms), pre- and post-injection of 0.15 mmol/kg Gd-DOTA. The pre-contrast images are at $t = 0$. The curve has been least-square-fit with the double exponential from Eq. (1).

curve has been least-square-fitted with the double exponential from Eq. (1).

Figure 3 illustrates the infinite cluster formed after determining the “with flow”/“no flow” pixels, and after running the Hoshen-Kopelman algorithm. For this example, there is a total number of 2490 pixels, the concentration of the cluster is $p = 0.3924$, and the lattice dimension is 58×58 pixels ($\sim 36 \times 36$ mm²). For this specific case, the fractal characteristics of the cluster are depicted in Fig. 4(a), where $d_f = 1.6962 \pm 0.011$; $\xi = 8.9283$; $p_\infty = 0.537$. The log-log plot of the “mass” of the cluster versus the length scale is also shown in Fig. 4(b). This plot is typical for reporting the fractal range scale. The fractal dimension of the minimum path was computed to be $d_{\min} = 1.086$. Figure 5(a, b) summarizes the values of d_f , $d_{\min}$ (Fig. 5(a)), and ξ (Fig. 5(b))

for the canine cases studied, as a function of the cluster concentration p .

Given that the backbone of the percolation cluster is responsible for the transport phenomena in the cluster, the “burning” algorithm was applied on the cluster in Fig. 3. Figure 6 shows the backbone (in darker gray) and the eliminated dangling ends (white). The backbone is almost identical with the initial cluster ($p_B = 0.3843$), its fractal dimension is $d_{f,B} = 1.6908 \pm 0.01$, the correlation length is $\xi_B = 9.69$, and $d_{\min,B} = 1.083$. Figure 5(a) also shows the fractal dimension of the backbone for the cases studied for an easy comparison with the values obtained for the whole clusters.

Study Two. The seed of each cluster was determined as the location of the maximum RPI⁵ value for all twenty two-dimensional tumor images in Study One. The clusters were then grown following the algorithm described in the “Materials and Methods” Section of Study Two. Figures 7(a–d) show the cluster grown after respectively 25, 50, 75, and 100 percent from the total growing time for the same case as illustrated in Fig. 2. The cluster in Fig. 7(d) is the final cluster obtained for Study One. For each intermediate cluster, the fractal dimension was computed as, respectively, $d_f^{25\%} = 1.50$ at a concentration $p_{25\%} = 0.1$; $d_f^{50\%} = 1.542$ at $p_{50\%} = 0.196$, $d_f^{75\%} = 1.63$ at $p_{75\%} = 0.324$, and $d_f^{100\%} = 1.6962$ at $p_{100\%} = 0.3924$. Similar simulations were performed for all data considered in Study One. For exemplification, results for the midplanes of each canine studied are summarized in Table 1.

Discussion and Conclusion

Although the tumor vascular network can be depicted rigorously only in three dimensions, several two-dimensional cases were studied to draw some general preliminary conclusions and to validate the methodology.

The contrast wash-in rate (i.e., tumor perfusion) data can be described using percolation models. The MRI images of the tumors were regarded as two-dimensional lattices and the scaled

$$^5 \text{RPI} = S_{\max} r^1.$$

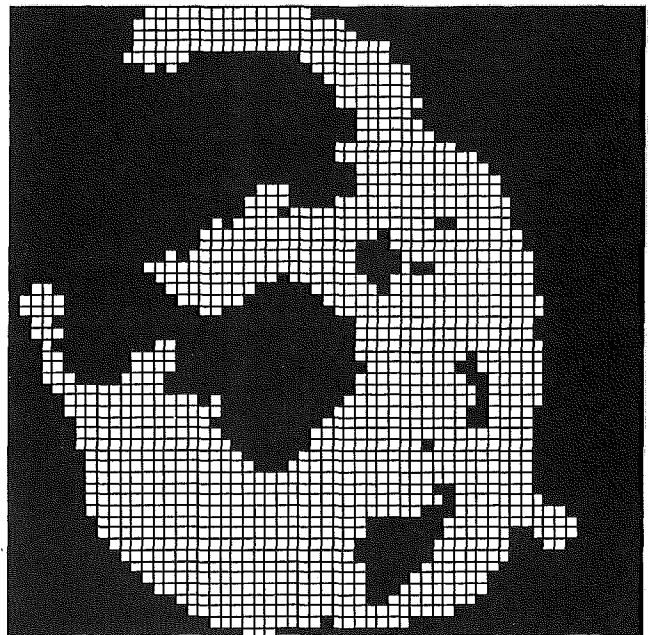


Fig. 3 Percolation cluster depicted from Fig. 2 by assuming pixels “with flow,” those with wash-in greater than the minimum wash-in, and by removing all the finite clusters. The size of the lattice is 58×58 pixels ($\sim 36 \times 36$ mm²).

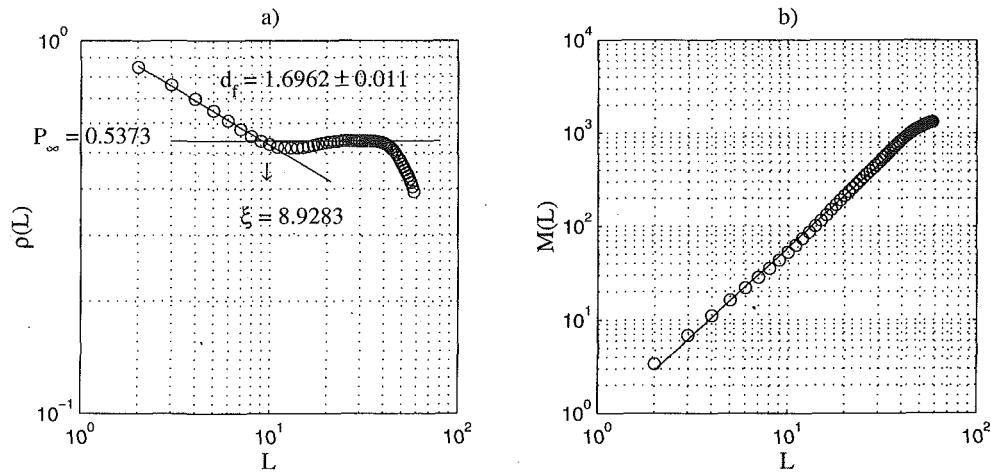


Fig. 4(a) Average density versus lattice dimension for the cluster in Fig. 3. Three regions can be noticed, as typical for percolation clusters (Kapitulnik et al., 1983): region I follows a power law, $\rho(L) \propto L^{d-d_f}$; region II is independent of L and identified as P_∞ , and region III shows boundary effects. (b) Average mass versus lattice dimension for the cluster in Fig. 3 to show the fractal scale range obtained for this particular case: 1.2 decades

values of the contrast wash-in were used to map the perfusion front pixel by pixel. A percolation model was then used to build this structure in terms of occupied and unoccupied sites, i.e., regions “with flow” versus regions with “no flow.” After the percolating cluster was isolated, the fractal characteristics of the cluster were determined using a two-dimensional analysis.

The three regions in Fig. 4(a), typical to percolation clusters, were obtained for all the cases studied, leading to the conclusion that the blood perfusion in a two-dimensional tumor can be described using a percolation model. Summarizing the results obtained from several canine tumors, Figs. 5(a, b) show the values obtained for the fractal dimension, fractal dimension of the minimum path, fractal dimension of the backbone, and respectively, the correlation length. It is worth mentioning that the fractal dimensions were obtained on scaling ranges spanning between 1.1–1.5 decades (1.2 decades for the example case shown in Fig. 4(b)). The lower cutoff was dictated by the size of the pixel obtained with the DE-MRI technique, whereas the upper cutoff was determined by the size of the structure (the tumor size). This limited range of length scales is typical for weakly correlated systems (percolation clusters under the percolation threshold), which was shown to exhibit fractal behavior over 0.5–2 decades (Malcai et al., 1997; Lidar et al., 1997).

Due to the fact that blood perfusion is a transport process, the backbone of the percolation cluster was isolated for each of the cases studied. From the data shown in Figs. 5(a) and 6 it is concluded that the fractal dimension of the backbone is on average 0.19 percent lower than the fractal dimension of the percolation cluster. The small differences between the concentration of the backbone and the percolation cluster may appear only in the two-dimensional analysis where there is no connectivity in the third direction, whereas in three dimensions, the backbone and the percolating cluster may be essentially the same (to be proven on three-dimensional geometries). The previous data showed that the percolation cluster and the backbone are essentially the same.

The invasion percolation growth model was used to grow the perfusion front in time. Using the invasion percolation growth model the perfusion front was then grown for all the cases studied. The front was allowed to grow only in sites where the wash-in value was greater than the minimum value, i.e., only in pixels “with flow.” This restriction for the invasion percolation front is justified by the existing wash-in data that characterizes perfusion. For the example shown in Fig. 7(a–c) it can be seen that the perfusion front develops initially along the edge of the tumor, penetrating the necrotic core only to a small extent. From the data in Table 1, it can be concluded that the evolution in time of the

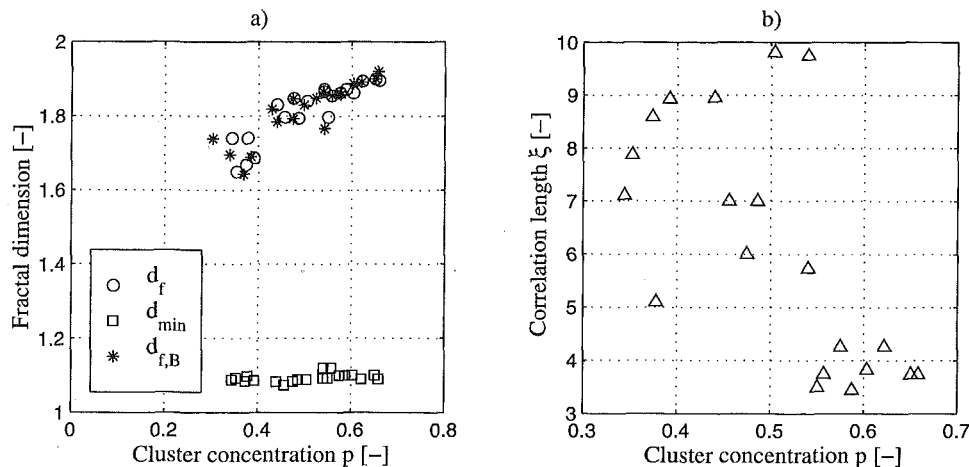


Fig. 5(a) Distribution of the fractal dimensions d_f , $d_{\min}$, and $d_{f,B}$ for different cluster concentrations p for all 20 structures studied. On average, the fractal dimension of the backbone is 0.19 percent less than that of the percolation cluster. (b) Distribution of the correlation length for different cluster concentrations p for all 20 structures studied

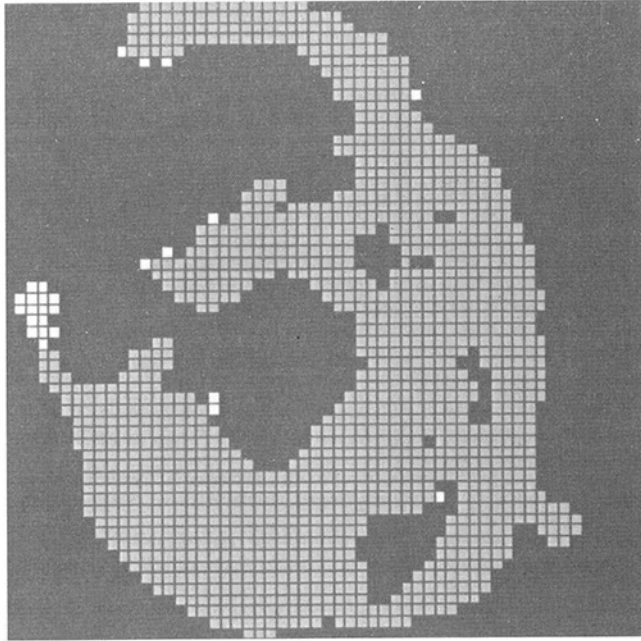


Fig. 6 Backbone of the percolation cluster from Fig. 3 (gray) and the removed dangling ends (white): $(p - p_c)/p = 0.005$

perfusion front is fractal as well. Also, after the growth was terminated (Fig. 7(d)) the structure obtained was the same as the one depicted in the wash-in images (see Figs. 2 and 3). Although the fractal dimensions of these two-dimensional clusters do not perfectly match the values reported in the literature for two-dimensional invasion percolation, the fact that by growing the perfusion front using the invasion percolation growth model the initial structure was obtained, led to the conclusion that the dynamics of blood perfusion in the tumor vascular network can be described by invasion percolation. The reconstruction of the three-dimensional cluster from two-dimensional slices will give the final answer with respect to the fractal dimension.

The number of iterations needed to grow the perfusion front completely is a function of the cluster concentration p and the tumor

Table 1 Values of the cluster concentration and fractal dimension for the intermediary clusters grown by invasion percolation, and for the clusters obtained from the wash-in data (initial clusters). The cases summarized represent midplane clusters of all six canines studied.

Case no.	25%		50%		75%		100%		Initial cluster	
	p	d_f	p	d_f	p	d_f	p	d_f	p	d_f
1	0.162	1.51	0.315	1.67	0.467	1.91	0.622	1.89	0.622	1.89
2	0.135	1.51	0.270	1.63	0.400	1.76	0.534	1.89	0.539	1.87
3	0.164	1.47	0.326	1.83	0.493	1.88	0.657	1.89	0.657	1.89
4	0.142	1.47	0.276	1.65	0.414	1.74	0.55	1.797	0.55	1.80
5	0.090	1.40	0.190	1.56	0.280	1.68	0.378	1.8	0.378	1.74
6	0.113	1.56	0.220	1.59	0.330	1.73	0.439	1.837	0.475	1.85

tortuosity reflected in the magnitude of $d_{\min}$. The fractal dimension of the minimum path is related to physical phenomena that propagate efficiently from site to site (Bunde and Havlin, 1991). The efficiency is related to the fact that a site can not be re-visited. We assumed that this is the case in a perfused tumor. In fractal theory, at cluster concentrations close to the critical value p_c , $d_{\min}$ characterizes also the velocity with which the front is propagated. From Fig. 5(a) one might draw the conclusion that $d_{\min}$ is almost constant for any cluster concentration, but this will imply a loss of useful information. A precise value of $d_{\min}$ is important in our application. The velocity of the moving front (perfusion front) scales⁶ in time with the exponent $1 - d_{\min}$. Since $d_{\min}$ is numerically close to 1 (for two dimensions), even a small error in $d_{\min}$ can have a large effect on the velocity component (Herrmann and Stanley, 1988). $d_{\min}$ can be a direct measure of the perfusion and, therefore, closely related to the tumor heatability. Thus, future work will be to monitor different tumors with different degrees of perfusion and test if there is a relationship between $d_{\min}$ and heatability.

In conclusion, the two-dimensional analysis shows a fractal behavior of the tumor perfusion front. The results from both studies presented lead to the conclusion that the tumor perfusion front can be described by the invasion percolation model. The two-dimensional analysis of the perfusion front growth shows that the growth is governed by the principles of invasion percolation. The growth mechanism is important in order to determine an exact value for $d_{\min}$, which is a parameter related to the perfusion front velocity, i.e., the tumor heatability. For all the cases, the front starts to grow from the edge of the tumor (where the feeding vessel was assumed to lie) and continues to grow on the periphery of the tumor, accounting for the well-documented perfused periphery and necrotic core of the tumor tissue.

The two-dimensional analysis was only performed to gain insight in the feasibility of applying percolation models in the description of the tumor perfusion front. After reconstructing the three-dimensional tumor blood perfusion from two-dimensional DE-MRI data, the final conclusions regarding the fractal structure will be drawn, i.e., the best statistical growth method that fits the experimental data.

Acknowledgments

The authors wish to thank Dr. J. Poulson and her collaborators for the DE-MRI dog data. This work was supported by NIH grants R01CA685129 and P01CA42745.

References

- Baish, J. W., Gazit, Y., Berk, D. A., Nozue, M., Baxter, L. T., and Jain, R. K., 1996, "Role of Tumor Vascular Architecture in Nutrient and Drug Delivery: An Invasion Percolation-Based Network Model," *Microvascular Research*, Vol. 51, pp. 327–346.
- Belfi, C. A., Ting, L. L., Hassenbusch, S. J., Tefft, M., and Ngo, F. Q., 1992, "Determination of Changes in Tumor Blood Perfusion After Hydralazine Treatment by Dynamic Paramagnetic-Enhanced Magnetic Resonance Imaging," *Int. J. Rad. Onc., Biol. Phys.*, Vol. 22, No. 3, pp. 477–482.
- Belfi, C. A., Paul, C. R., Shan, S., and Ngo, F. Q., 1994, "Comparison of the Effects of Hydralazine on Tumor and Normal Tissue Blood Perfusion by MRI," *Int. J. Rad. Onc., Biol. Phys.*, Vol. 29, No. 3, pp. 473–479.

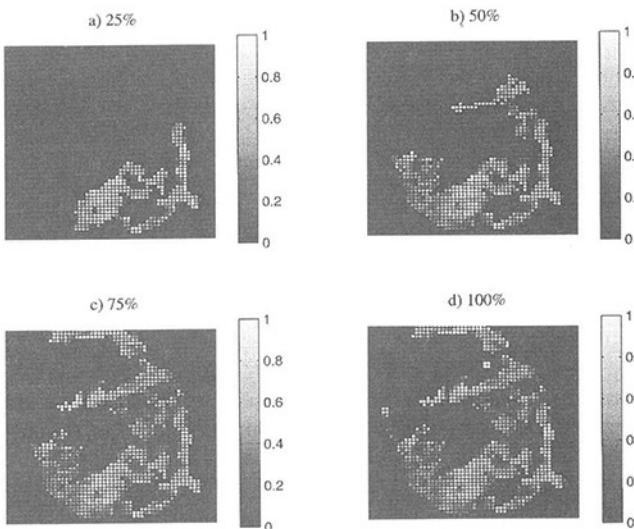


Fig. 7 Clusters grown by invasion percolation starting from the seed, shown after: (a) 25, (b) 50, (c) 75, and (d) 100 percent of the total growing time

⁶ $v = dr/dt = dr/dl_c = (dl_c/dr)^{-1} = r^{1-d_{\min}} = (p - p_c)^{v(d_{\min}-1)}$, where $t = l_c$ is the minimum path length between the newest site added and the origin (Stanley, 1991, p. 35).

- Binder, K., and Heerman, D. W., 1992, *Monte Carlo Simulations in Statistical Physics*, Springer-Verlag.
- Bunde, A., and Havlin, S., 1991, "Percolation I and II," in: *Fractals and Disordered Systems*, Bunde and Havlin, eds., Springer-Verlag, pp. 51–150.
- Bunde, A., and Havlin, S., 1994, "A Brief Introduction to Fractal Geometry," in: *Fractals in Science*, Bunde and Havlin, eds., Springer-Verlag, pp. 1–25.
- Craciunescu, O., 1998, "Influence of Blood Vessel Networks on Hyperthermia Induced Temperature Distributions," Ph.D. Thesis, Duke University, pp. 9–11.
- Daxter, A., and Ettl, A., 1995, "Corneal Vascularization and Its Relation to the Physical Properties of the Tissue: A Fractal Analysis," *Current Eye Research*, Vol. 14, No. 4, pp. 263–268.
- Daxter, A., 1993, "The Fractal Geometry of Proliferative Diabetic Retinopathy: Implications for the Diagnosis and the Process of Retinal Vasculogenesis," *Current Eye Research*, Vol. 12, No. 12, pp. 1103–1109.
- Dewhirst, M. W., 1993, "Angiogenesis and Blood Flow in Solid Tumors," in: *Drug Resistance in Oncology*, B. Teicher, ed., Marcel Dekker, Inc., New York, pp. 3–24.
- Dewhirst, M. W., Prosnitz, L., Thrall, D., Prescott, D. M., Clegg, S. T., Charles, H. C., MacFall, J. R., Rosner, G., Samulski, T. V., Gillette, E., and LaRue, S., 1997, "Hyperthermic Treatment of Malignant Diseases: Current Status and a View Toward the Future," *Seminars in Oncology*, Vol. 24, No. 6, pp. 616–625.
- Feder, J., 1988, *Fractals*, Plenum Press, New York.
- Furunberg, L., Feder, J., Aharony, A., and Jøssang, T., 1988, "Dynamics of Invasion Percolation," *Physical Review Letters*, Vol. 61, No. 18, pp. 2117–2120.
- Gazit, Y., Berk, D. A., Leuning, M., Baxter, L. T., and Jain, R. K., 1995, "Scale-Invariant Behavior and Vascular Network Formation in Normal and Tumor Tissue," *Physical Review Letters*, Vol. 75, No. 12, pp. 2428–2431.
- Gazit, Y., Baish, J., Safabakhsh, N., Leuning, M., Baxter, L. T., and Jain, R. K., 1997, "Fractal Characteristics of Tumor Vascular Architecture During Tumor Growth and Regression," *Microcirculation*, Vol. 4, No. 4, pp. 395–402.
- Gebele, T., 1984, "Site Percolation Threshold for a Square Lattice," *J. Phys. A: Math. Gen.*, Vol. 17, pp. L51–54.
- Herrmann, H. J., Hong, D. C., and Stanley, H. E., 1984, "Backbone and Elastic Backbone of Percolation Clusters Obtained by the New Method of 'Burning'," *J. Phys. A: Math. Gen.*, Vol. 17, pp. L261–266.
- Herrmann, H. J., and Stanley, H. E., 1988, "The Fractal Dimension of the Minimum Path in Two- and Three-Dimensional Percolation," *J. Phys. A: Math. Gen.*, Vol. 21, L829–L833.
- Hoshen, J., and Kopelman, R., 1976, "Percolation and Cluster Distribution. I. Cluster Multiple Labeling Technique and Critical Concentration Algorithm," *Physical Review B*, Vol. 14, No. 8.
- Hudetz, A. G., 1993, "Percolation Phenomenon: The Effect of Capillary Network Rarefaction," *Microvascular Research*, Vol. 45, pp. 1–10.
- Hudetz, A. G., and Werin, S., 1986, "Percolation and Transit in Microvascular Networks," *Adv. Exp. Med. Biol.*, Vol. 200, pp. 74–88.
- Kapitulnik, A., Aharony, A., Deutscher, G., and Stauffer, D., 1983, "Self Similarity and Correlation in Percolation," *J. Phys. A: Math. Gen.*, Vol. 16, pp. L269–L274.
- Kiani, M. F., and Hudetz, A. G., 1991, "Computer Simulation of Growth of Anastomosing Microvascular Networks," *J. Theor. Biol.*, Vol. 150, pp. 547–560.
- Kurland, R. J., Shoop, J. D., and Funkhouser, G. R., 1989, "A Kinetic Model for Time Development of Gd-DTPA in Relaxation Enhancement (Abstr.)," *Magn. Reson. Imag. (Suppl.)*, Vol. 177.
- Lidar (Hamburger), D. A., Biham, O., and Avnir, D., 1997, "Limited Range Fractality of Randomly Absorbed Rods," *J. Chem. Phys.*, Vol. 106, p. 10359.
- Malcai, O., Lidar, D. A., Biham, O., and Avnir, D., 1997, "Scaling Range and Cutoffs in Empirical Fractals," *Phys. Rev. E*, Vol. 56930, pp. 2817–2828.
- Mandelbrot, B. B., 1982, *The Fractal Geometry of Nature*, Freeman, New York.
- Mastorakos, J., and Argyrakis, P., 1993, "Transport on the Percolation Backbone," *Phys. Rev. E*, Vol. 48, No. 6, pp. 4847–4850.
- McCarthy, J. F., 1987, "Invasion Percolation on a Random Lattice," *J. Phys. A: Math. Gen.*, Vol. 20, pp. 3465–3469.
- Pennes, H. H., 1948, "Analysis of Tissue and Arterial Blood Temperatures in Resting Human Forearm," *J. Appl. Physiology*, Vol. 1, pp. 93–122.
- Rinck, P. A., ed., 1993, "Contrast Agents," in: *Magnetic Resonance in Medicine: The Basic Textbook of the European Magnetic Resonance Forum*, 3rd ed., Blackwell Scientific Publications, pp. 155–174.
- Samulski, T. V., and Fessenden, P., 1990, "Thermometry in Therapeutic Hyperthermia," in: *Methods of Hyperthermia Control*, Gautherie, M., ed., Springer-Verlag, pp. 1–34.
- Stanley, E. H., 1991, "Fractals and Multifractals: The Interplay of Physics and Geometry," in: *Fractals and Disordered Systems*, A. Bunde and S. Havlin, eds., Springer-Verlag, pp. 1–49.
- Stauffer, D., and Aharony, A., 1992, *Introduction to Percolation Theory*, Taylor & Francis.
- Tsonis, A. A., and Tsonis, P. A., 1987, "Fractals: A New Look at Biological Shape and Patterning," *Perspectives in Biology and Medicine*, Vol. 30, No. 3, pp. 355–361.
- Vico, P. G., Boyer, H., Cartilier, L. H., 1992, "New Concepts in the Study of Tissue Vascularization: A Mathematical Model of Skin Vascularization," *Plastic and Reconstructive Surgery*, Vol. 94, No. 1, pp. 174–179.
- Vicsek, T., 1989, *Fractal Growth Phenomena*, World Scientific.
- Warren, B. A., 1979, "The Vascular Morphology of Tumors," in: *Tumor Blood Circulation: Angiogenesis, Vascular Morphology, Blood Flow of Experimental and Human Tumors*, H. I. Peteresen, ed., CRC Press.
- Wilkinson, D., and Willemsen, J. F., 1983, "Invasion Percolation: A New Form of Percolation Theory," *J. Phys. A: Math. Gen.*, Vol. 16, pp. 3365–3376.
- Wilkinson, D., and Barsony, M., 1984, "Monte Carlo Study of Invasion Percolation Clusters in Two and Three Dimensions," *J. Phys. A: Math. Gen.*, Vol. 17, pp. L129–L135.